

SAP Convergent Invoicing (CI) Consumption & Billable Item Management Master Guide

Product: SAP Convergent Invoicing

Module: CI / BRIM

Audience: Senior SAP BRIM Enterprise Architects & Consultants (8+ Years Experience)

Classification: Enterprise Official Technical Handbook

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Section 1: Convergent Invoicing Architectural Overview & Core Workflow

SAP Convergent Invoicing (CI) bridges the gap between rating/charging engines (SAP CC, external rating) and financial subledgers (FI-CA, General Ledger). CI consolidates high-volume usage, one-off, and recurring charges into clean customer invoices. Core flow:

1. Consumption Items (CIT): Ingestion of unrated usage events.
2. Rating: Translation of CITs into rated monetary charges.
3. Billable Items (BIT): Management of rated charges across status lifecycles.
4. Billing (FKKBIX_BILL_M): Aggregation of BITs into Billing Documents.
5. Invoicing (FKKBIX_INV_M): Integration of billing documents, open items, taxes, and creation of Invoicing Documents (ERDK) and FI-CA items (DFKKOP).

Reference: SAP BRIM Technical Standard - CI / BRIM - Section 1

Section 2: Consumption Item (CIT) Classes & Management (FKKBIX_CIT_CONF)

Consumption Item Classes define the technical and business attributes of raw usage data:

- Configuration: Executed via transaction FKKBIX_CIT_CONF.
- Table Generation: The system generates physical database tables for each class: DFKKBIXCIT0 (Raw unrated items), DFKKBIXCIT2 (Rated items).
- Ingestion Interfaces: High-performance RFC (BAPI_CTRAC_CONSUMPTION_ITEM), Web Services, or mass file upload (FKKBIX_CIT_UPLOAD).
- Field Categories: Technical key fields (source transaction ID, timestamp), Consumption fields (duration, volume, unit of measurement), and Rating control fields.

Reference: SAP BRIM Technical Standard - CI / BRIM - Section 2

Section 3: Billable Item (BIT) Classes & Lifecycle Status Architecture

Billable Item Classes structure rated monetary charges ready for customer billing:

- Configuration: Defined via transaction FKKBIX_BIT_CONF in SPRO.
- Status Architecture:
 - * Status 0: Raw (Table DFKKBIXBIT0) - Initial staged items awaiting validation.
 - * Status 1: Raw with Errors (Table DFKKBIXBIT1) - Validation failed (e.g. missing contract account, invalid transaction code).
 - * Status 2: Billable (Table DFKKBIXBIT2) - Validated, billable items ready for billing aggregation.
 - * Status 3: Billed (Table DFKKBIXBIT4) - Items successfully incorporated into a billing document.
 - * Status 4: Simulated (Table DFKKBIXBIT_SIM) - Used for bill simulation and forecasting.
- Mass Creation: Executed via FKKBIX_BIT_CREATE or direct API integration.

Reference: SAP BRIM Technical Standard - CI / BRIM - Section 3

Section 4: Billable Item Types, Subtypes, and Selection Variants

Categorizing and filtering billable items for targeted billing runs:

- Billable Item Type: Classifies charges into business categories such as Usage Charges, Recurring Subscription Fees, One-off Activation Fees, Minimum Spend Surcharges, and Early Termination Penalties.
- Billable Item Subtype: Provides granular sub-categorization for reporting and pricing differentiation.
- Selection Variants: Configuration rules that filter specific BIT types and contract accounts for tailored billing cycles (e.g. monthly vs quarterly billing runs).

Reference: SAP BRIM Technical Standard - CI / BRIM - Section 4

Section 5: Billing Process Mass Activity (FKKBIX_BILL_M)

The Billing Process aggregates billable items into official Billing Documents:

- Transaction: FKKBIX_BILL_M executes mass billing across parallel application server intervals.
- Billing Document Architecture:
 - * Header Table: FKKBIX_BILLDOC_H (Billing document number, contract account, billing date, total amounts).
 - * Item Table: FKKBIX_BILLDOC_IT (Aggregated billing lines, main/subtransactions, tax codes, period coverage).
 - * Tax Table: FKKBIX_BILLDOC_TX (Calculated tax lines).
- Billing Orders: Scheduling triggers that manage when contract accounts are due for billing.
- Billing Reversal: Transaction FKKBIX_BILL_REV allows reversing billing documents, returning BITs to Billable status (Status 2).

Reference: SAP BRIM Technical Standard - CI / BRIM - Section 5

Section 6: Invoicing Process Mass Activity (FKKBIX_INV_M)

Invoicing is the culminating financial mass run in SAP BRIM:

- Transaction: FKKBIX_INV_M executes mass invoicing across interval selections.
- Invoicing Document Architecture: Header table ERDK, item table ERDZ.
- Core Invoicing Functions:
 1. Aggregation of Billing Documents: Pulls billing documents from CI and external billing sources.
 2. Open Item Clearing: Applies clearing control (Posting Area 0150) to clear existing FI-CA credits against new charges.
 3. Tax Determination: Calculates statutory sales taxes, VAT, or integrates with third-party tax engines.
 4. Creation of FI-CA Subledger Items: Posts open receivables into DFKKOP.
 5. Print Document Generation: Prepares print records for bill layout formatting.
- Invoicing Reversal: Transaction FKKBIX_INV_REV reverses invoicing documents and un-clears associated FI-CA open items.

Reference: SAP BRIM Technical Standard - CI / BRIM - Section 6

Section 7: Convergent Invoicing Integration with SAP RAR (IFRS 15 / ASC 606)

Compliance with statutory revenue recognition standards:

- Revenue Accounting Items (RAI): CI generates RAIs representing order items, fulfillment items, and invoice items.
- RAI Classes: Configured in CI to capture standalone selling prices (SSP) and performance obligations (POB).
- RAI Monitor (FARR_RAI_MON): Administrative transaction to monitor, validate, and transfer RAIs from Convergent Invoicing to the SAP RAR engine.
- Contract Modifications & Reversals: CI coordinates invoice reversals with RAR contract revisions to adjust deferred revenue balances.

Reference: SAP BRIM Technical Standard - CI / BRIM - Section 7

Section 8: Exception Handling, BIT Monitoring, and Clarification Processing

Senior operational monitoring tools for CI health:

- Transaction FKKBIX_BIT_EXCP: Inspects and corrects billable items in error status (Status 1).
- Error Analysis: Common causes include unassigned main/subtransactions, missing provider contracts, and locked billing accounts.
- Reprocessing: Corrected items are re-validated and promoted to Status 2 (Billable) without data loss.
- Mass Analysis Reports: RFKKBIX_BIT_DISPLAY provides comprehensive filtering and auditing of item states.

Reference: SAP BRIM Technical Standard - CI / BRIM - Section 8

Section 9: BAdI Extension Architecture in Convergent Invoicing

Key Business Add-Ins (BAdIs) for custom logic implementation:

- FKKBIX_BIT_PREPARE: Custom enrichment and validation during billable item upload.
- FKKBIX_BILL_CREATE: Intercepts billing document creation to apply custom discount calculations or customer grouping.
- FKKBIX_INVOICING: Modifies invoicing document items (ERDZ), print layouts, or triggers external event notifications upon bill finalization.
- Performance Rule: BAdI logic must be optimized for batch memory; database calls inside row loops are strictly prohibited.

Reference: SAP BRIM Technical Standard - CI / BRIM - Section 9

Section 10: CI Performance Optimization and Parallelization Tuning

Maximizing mass billing and invoicing throughput:

- Interval Tuning: Balance interval count with available background work processes (SM50/SM66).
- Commit Frequency: Configure optimal commit counters in billing process customizing to prevent lock table overflow.
- Table Archiving: Periodically archive processed billable items (DFKKBIXBIT4) using archiving object FKKKBIXBIT to preserve query performance.

Reference: SAP BRIM Technical Standard - CI / BRIM - Section 10